Q- Sample-1: Easy

Given:- $n = 5$, $A = [5, 5, 0, 3, 0]$.
$q = 5$. queries $= [[0, 2, 1, 2], [0, 1, 6, 5], [2, 3, 8, 0], [2, 4, 9, 6], [3, 4, 8, 9]]$

for query 1:-

$\Rightarrow l = 0, r = 2, x = 1, y = 2$.

- $A[0] = 1$ $\{A[l] = x\}$
- $A[1] = 3$ $\{A[l+1] = x + y\}$
- $A[2] = 5$. $\{A[l+i] = x + i * y$, where $i = 0 \to (r-l)$
  $A[l+2] = x + 2 * y\}$

So, $A = [1, 3, 5, 3, 0]$.

$\Rightarrow$ for query 2:-

$l = 0, r = 1, x = 6, y = 5$.

- $A[0] = 6$
- $A[1] = 11$

So, $A = [6, 11, 5, 3, 0]$.

$\Rightarrow$ for query 3:-

$l = 2, r = 3, x = 8, y = 0$.

- $A[2] = 8$
- $A[3] = 8$.

So, $A = [6, 11, 8, 8, 0]$

$\Rightarrow$ for query 4:-

$l = 2, r = 4, x = 9, y = 6$.

- $A[2] = 9$
- $A[3] = 15$
- $A[4] = 21$.

So, $A = [6, 11, 9, 15, 21]$

$\Rightarrow$ for query 5:-

$l = 3, r = 4, x = 8, y = 9$.

- $A[3] = 8$.
- $A[4] = 17$

So, $A = [6, 11, 9, 8, 17]$

Hence, answer is $6 + 11 + 9 + 8 + 17 = 51$

# BRUTE FORCE:-

⇒ Approach:- Using AP:-

$A[l+i] = x + i*y$, where $i = 0 \to (r-1)$

→ Value = x.
→ for l → r.
  → A[i] = value
  → value = value + y. (AP)
  → sum (A)

⇒ Pseudocode:-

```
mod = 10**9 + 7 ( mod = 1000000007).
def solve (n, A, queries):
    for l, r, x, y in queries:
        value = x.
        for i in range(l, r+1):
            A[i] = value.
            value += y.

    return sum(A)%mod.
```

⇒ TC:- O(q*n)
SC:- O(1)

⇒ Dry Run:-

A = [5, 5, 0, 3, 0];

(1.) q1 ⇒ l = 0, r = 2, x = 1, y = 2.
value = 1.
- i = 0 → A[0] = 1.
  value = 3.
- i = 1 → A[1] = 3.
  value = 5
- i = 2 → A[2] = 5
  value = 7.

A = [1, 3, 5, 3, 0].

(2.) q2 ⇒ l = 0, r = 1, x = 6, y = 5
value = 6.
- i = 0 → A[0] = 6.
  value = 11
- i = 1 → A[1] = 11
  value = 16.

A = [6, 11, 5, 3, 0]

(3.) q3 ⇒ l = 2, r = 3, x = 8, y = 0.
value = 8
- i = 2 → A[2] = 8
  value = 8
- i = 3 → A[3] = 8
  value = 8

A = [6, 11, 8, 8, 0].